RESEARCH BRIEFING

Explainable Cyber Threat Defense Workbench

Comparing readable detection signals and interpretable models on UNSW-NB15 network flows

NETWORK-FLOW ATTACK DETECTION · ONE-NOTEBOOK MVP

Class project · October 2026

Executive readout

The benchmark separates model ranking from threshold transfer and review workload.

73,791

primary test flows

after removing exact train/test input overlap

51.9%

attack-labeled

positive-label share in the primary test

0.9825

XGBoost test AP

highest average precision among the three models

EQUAL-SIZE REVIEW COMPARISON

27,973

flows reviewed by each approach

59.0%

rule-union recall

72.9%

XGBoost recall at same alert count

READOUT Strong ranking performance on this split; very high benchmark precision; validation cutoff flagged far fewer test flows than expected.

Detection patterns evolved by adding context and learning

These approaches overlap in practice; each contributes a different kind of evidence.

01 · MATCH KNOWN CONDITIONS

Rules and signatures

Recognize specified indicators or flow conditions.

02 · FIND DEVIATIONS

Statistical anomaly detection

Flag behavior that departs from an established baseline.

03 · LEARN FROM LABELS

Supervised flow scoring

Rank traffic using examples labeled benign or malicious.

04 · COMBINE EVIDENCE

Hybrid review queues

Fuse signals and prioritize alerts within review capacity.

EVOLUTION

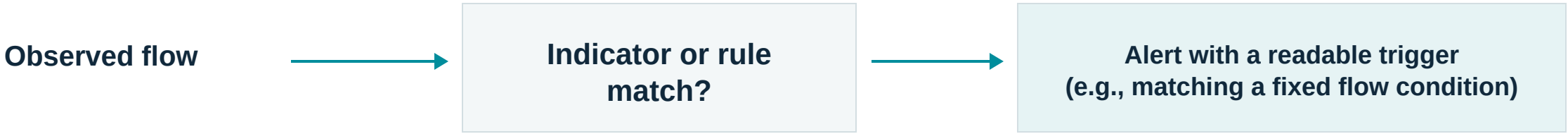
Detection expanded from explicit matches toward behavioral context, learned ranking, and combined evidence. Earlier patterns remain useful alongside newer ones.

Sources: Denning (1987), DOI 10.1109/TSE.1987.232894 · Roesch (1999), USENIX LISA

Pattern 1: signatures and rules match specified behavior

This pattern is direct and inspectable: a known indicator or condition triggers an alert.

HOW IT WORKS



STRENGTHS

Clear trigger; fast to explain and investigate; effective when the behavior is known and the condition stays relevant.

TRADE-OFF

Coverage depends on the rules and indicators already encoded; new or changing behavior can evade fixed matches.

IN THIS PROJECT

Three simple rules create explicit alert flags that later become model inputs.

Reference: Roesch (1999), Snort—Lightweight Intrusion Detection for Networks, USENIX LISA.

Pattern 2: anomaly detection asks what departs from a baseline

Instead of matching a named attack, the detector scores unusual behavior relative to learned normal activity.

NORMAL ACTIVITY

Build a profile of expected behavior.

→

DEVIATION

Measure how far a flow departs from that profile.

→

ANOMALY SCORE

Prioritize unusual records for investigation.

WHY IT ADDED A NEW LENS

A deviation score can surface behavior that lacks a hand-written signature, but unusual does not automatically mean malicious.

Reference: Denning (1987), An Intrusion-Detection Model, IEEE TSE, DOI 10.1109/TSE.1987.232894.

Pattern 3: supervised models learn from labeled flow examples

A classifier estimates attack likelihood from feature combinations represented in labeled data.

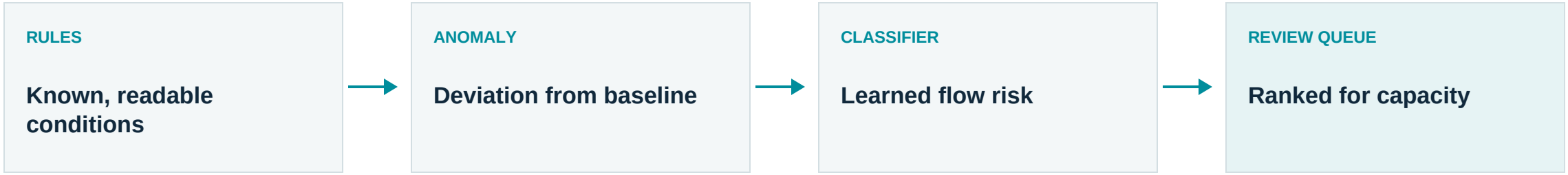
KEY DEPENDENCY

The model inherits the coverage and biases of the labels, features, and evaluation split used to train and assess it.

Dataset reference: Moustafa & Slay (2015), UNSW-NB15, DOI 10.1109/MilCIS.2015.7348942.

Pattern 4: hybrid detection combines signals for review

A layered workflow can retain explicit rules while adding anomaly and learned risk scores.



WHY COMBINE?

Different signals capture different evidence. A shared ranking or side-by-side queue can expose trade-offs between coverage, false alerts, interpretability, and review effort.

The central choice becomes operational: which signals to surface, how to rank them, and how much work the queue creates.

This is a conceptual design pattern; exact implementations and alert policies vary by system.

Project MVP: fuse readable signals into per-flow model scores

The project turns the historical pattern sequence into one compact, explainable comparison.

FLOW EVIDENCE

17 numeric fields
proto · service · state

→

EXPLICIT SIGNALS

3 rule flags
anomaly score
connection-count score

→

THREE MODEL OPTIONS

Logistic regression
XGBoost
Explainable Boosting Machine

→

REVIEW OUTPUT

One score per flow
ranked queue
explanations and misses

MVP QUESTION

Do fused flow features and earlier detection signals improve the useful ranking of attack-labeled records at a controlled alert workload?

Evaluate the standalone tracks separately, then compare models and signals at matched review volume.

Study population and evaluation split

Published train/test files are attack-heavy and contain repeated feature rows.

175,341

training flows

119,341 attack-labeled · 68.1%

82,332

published test flows

45,332 attack-labeled · 55.1%

PRIMARY TEST POPULATION

8,541

exact input matches removed

→

73,791

flows retained

51.9%

attack-labeled share

SPLIT BOUNDARY CSV files have no timestamps; validation is grouped-random, not a future-period test.

Selected columns describe flow behavior and recent context

All classifiers share the same compact feature set.

Feature group	Selected columns	Plain-language meaning
Flow volume and timing	dur · rate · spkts · dpkts · sbytes · dbytes	Duration, packet and byte counts, traffic rate
Packet behavior	sttl · dttl · sload · dload · sloss · dloss	Directional TTL, load, and loss measures
Recent connection counts	ct_srv_src · ct_dst_ltm · ct_src_dport_ltm · ct_dst_sport_ltm · ct_dst_src_ltm	Recent activity by service, source/destination, and ports
Categorical context	proto · service · state	Protocol, service, and connection state

ADDED SIGNALS 3 rule flags · Isolation Forest score · connection-concentration score

Feature and split controls address key leakage risks

These controls remove direct label fields and exact cross-split input matches.

Field or condition	Treatment	Reason
id	Excluded from model inputs	Row identifier
label	Binary target only	Never a feature
attack_cat	Used after scoring for missed-category review	Attack type is unavailable to classifiers
Exact test input overlap	8,541 rows removed from primary test	Avoid scoring exact feature copies of training rows

Remaining data concerns

74,301 training rows and 28,386 published test rows repeat earlier inputs within their files; 229 training input groups have conflicting labels.

Standalone tracks trade recall against alert volume

Each track has its own alert rule or validation-set score cutoff.

Track	Alerts	Attack-labeled	Normal-labeled	Precision	Recall
Rules union	27,973	22,578	5,395	80.7%	59.0%
Isolation Forest	5,889	4,039	1,850	68.6%	10.6%
Connection concentration	7,157	7,157	0	100.0%	18.7%

DATASET-SPECIFIC RESULT

The connection-count track precision reflects this primary test sample; validate it on other traffic before treating it as a general property.

Model selection and threshold choice answer different questions

XGBoost was selected by validation AP; its validation cutoff did not preserve alert rate on test.

MODEL SELECTION

Validation AP

0.9950

Highest among three model families

FROZEN VALIDATION CUTOFF

90th percentile of validation scores

1,831

test alerts · 4.8% attack recall

FIXED TEST REVIEW BUDGET (RANKING CHECK)

7,379

flows at a 10% test budget

19.3%

attack recall

A ranked queue with fixed size is a separate policy comparison from a frozen score threshold.

XGBoost leads the held-out model comparison

Model selection used validation AP; this table reports the primary test metrics.

Model	Average precision	ROC AUC	Brier score
Logistic regression	0.9696	0.9654	0.1088
XGBoost	**0.9825**	**0.9796**	**0.0929**
Explainable Boosting Machine	0.9746	0.9710	0.1049

HOW TO READ THESE METRICS

Average precision

Ranking quality across thresholds

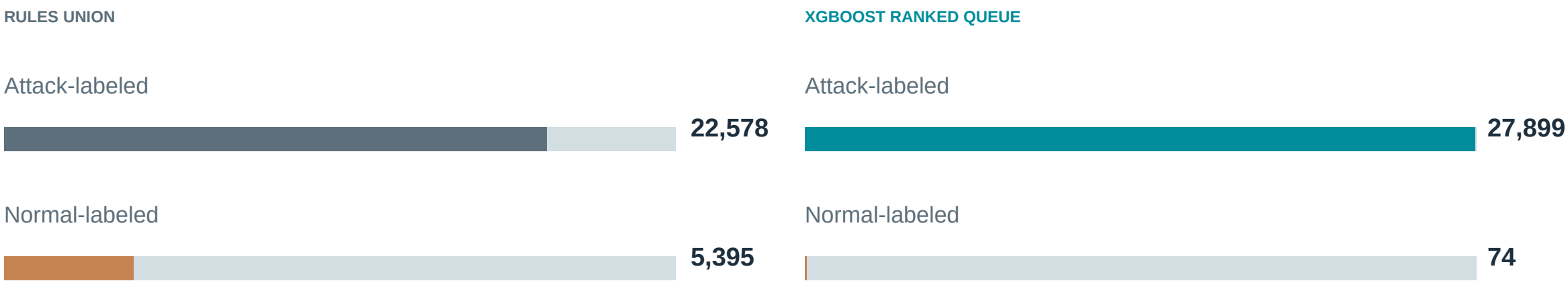
Brier score

Probability error; lower is better

The same primary test population is used for all three classifiers.

At the same alert volume, XGBoost finds more attack labels

Both queues contain 27,973 flows from the primary test set.



	RULES UNION	XGBOOST RANKED QUEUE
Precision	80.7%	99.7%
Recall	59.0%	72.9%

Matched workload isolates queue quality within this benchmark; it does not estimate production performance.

Supplementary tracks find some misses at added review cost

Counts are incremental beyond the equal-budget XGBoost queue.

Added track	Additional attack-labeled	Additional normal-labeled	Combined queue
Rule alerts	1,626	5,373	34,972 total alerts · 77.2% recall
Isolation Forest	857	1,830	Adds 857 attack-labeled flows
Connection concentration	0	0	No new alerts at this model budget

Additional attack-labeled

+1,626

flows

Additional normal-labeled

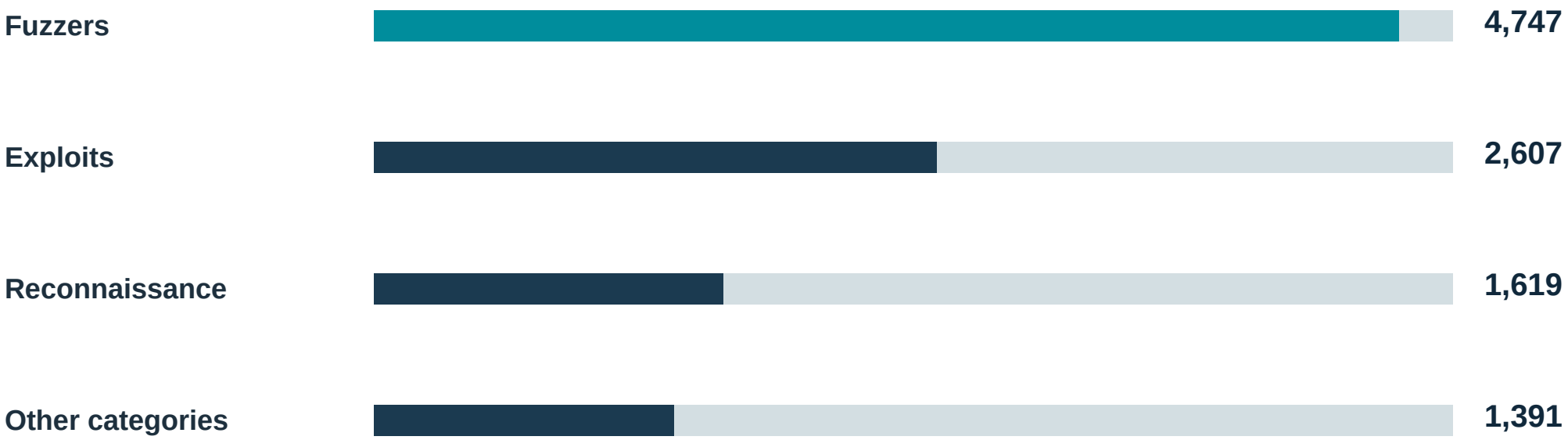
+5,373

flows

Combining tracks expands coverage and workload; this is a diagnostic comparison, not a selected operating policy.

Fuzzers, exploits, and reconnaissance account for the largest misses

At the rule-matched XGBoost alert budget, 10,364 attack-labeled flows remain unalerted.



Review question Which flow patterns and attack categories are absent from this queue, and what signals might surface them?

Model explanations describe scores and support review

Global and local views answer different questions.

Model	Explanation used in project	What it helps inspect
Logistic regression	Coefficients and one-hot contributions	Which inputs move the linear score
XGBoost	Global feature importance and local SHAP contributions	Which inputs matter overall and for a missed flow
Explainable Boosting Machine	Native global effects and local explanation	How learned feature effects add to a score

INTERPRETIVE LIMIT

A feature contribution explains the model output. It does not establish the cause of an attack or prove attacker intent.

The notebook shows one local explanation for a missed attack-labeled flow, alongside global views.

What the measured findings support

Evidence is strongest when the evaluation question is stated with the metric.

01 **Model ranking is strong on the primary test**

XGBoost leads AP, ROC AUC, and Brier score among the three classifier families.

02 **A matched queue favors XGBoost over the rules**

At 27,973 alerts, the model catches 5,321 more attack-labeled flows and sends 5,321 fewer normal-labeled flows for review.

03 **Threshold transfer needs attention**

The validation 90th-percentile cutoff produces 1,831 test alerts, while a fixed 10% ranking budget contains 7,379.

04 **The dataset pattern is a boundary**

Attack-heavy labels, repeated rows, and missing timestamps limit how far these results generalize.

These findings describe UNSW-NB15 benchmark behavior, not expected production performance.

What to validate before applying the workflow to other traffic

The next evidence should come from data and review conditions closer to the intended setting.

01 **Outcome labels**

Compare alerts with investigated incidents or documented ground truth.

02 **Time and network separation**

Evaluate later periods and networks; the published CSVs have no timestamps.

03 **Base rate and calibration**

Measure precision, false-alert volume, and calibration at realistic attack prevalence.

04 **Attack coverage**

Report misses by attack type, service, and traffic context.

05 **Review capacity**

Set alert budgets with analyst workflows and disposition feedback in view.

Sources and project record

Dataset citation, code, and saved findings.

Reference	Role in this briefing
Denning (1987)	Foundational statistical intrusion-detection model · DOI 10.1109/TSE.1987.232894
Roesch (1999)	Snort signature and rule-based network IDS · USENIX LISA paper
Moustafa and Slay (MilCIS 2015)	UNSW-NB15 dataset paper · DOI 10.1109/MilCIS.2015.7348942
UNSW-NB15 dataset page	Dataset release, published split, and citation guidance
Cyber Threat Defense Workbench repository	Notebook, setup instructions, project guide, and measured findings
Cyber Threat Defense Expert Briefing	Slide summary of history, data, methods, metrics, and interpretation

PROJECT FILES

cyber_threat_modeling.ipynb · docs/PROJECT_GUIDE.md · docs/FINDINGS.md

Dataset: https://research.unsw.edu.au/projects/unsw-nb15-dataset

Denning: https://doi.org/10.1109/TSE.1987.232894

Roesch: https://www.usenix.org/legacy/event/lisa99/full_papers/roesch/roesch.pdf